

Vision-Guided Robotic Sorting Cell

Progress report — 16 August 2026

The vacuum gripper fault that had halted the cell was traced not to damaged wiring, as first assumed, but to a single software call reading a register the tool board does not support; avoiding it removed the fault completely, from 148 failures in 296 operations to none in 297, with no hardware repair needed. Attention then moved to calibrating the camera to the arm, which was attempted twice and rejected both times: the first run measured 10.9 mm of error because the camera was averaging each cube's top face together with its side face, and the second measured 16.7 mm because guiding the arm by hand left the wrist at a different angle on every touch, when the method requires it to be vertical. Both causes have been identified and corrected — the measurement now isolates the top face alone, and the arm squares its own wrist and lowers itself until the vacuum sensor confirms contact — and the cell is ready for a third calibration run, after which the working surface will be taught and the first live pick attempted.